

ActInf GuestStream #031.1 ~ Brett Kagan & Adeel Razi

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foreign hello and welcome everyone it's December 7th 2022. this is active inference live stream number 31.1 we are here with Brett Kagan and we're going to be hearing a presentation and having a discussion on free energy principle and active inference in synthetic biological intelligence we'll have a presentation followed by a discussion so feel free to submit any questions if you would like Brett thank you for joining really looking forward to this presentation and discussion off to you thanks so much Daniel and thank you everyone for joining us today hopefully we've got some interesting stuff so what I want to talk to you about is the work that we've been doing uh in what we call synthetic biological intelligence and particularly the use we've made of uh principles from the free energy principle and active inference Frameworks so if we jump to sort of the first question and we can kind of ask ourselves because we work with neural systems what is there that's unique about neural systems and one of the obvious answers to that is that they display this unique ability to collate information and apply it in an Adaptive behavior in multiple contexts and that could be a fitting definition for intelligence if you like um but to actually be able to test this in actual sales in real time what you have to do is be able to record the information from the cells and provide information back to the cells in real time so how did we go about setting this up how did we get these cells so we did it in two ways either we took what's called a human induced pluripotent stem cell that's the hip SC there that you can see and what you can do is develop these from any donor blood tissue or skin and you can basically make a pluripotent stem cell line and then you can use a number of different methods to turn that into neurons of you know quite a degree of specificity we used quite a broad one for most of our work which is called a dual somat inhibition which follows like natural ontological development uh on antigenine development I should say uh to uh sort of create these predominantly cortical cultures uh in a dish but you can use other direct methods such as what's called an ngn2 direct differentiation which gives rise to a more excitatory culture uh then we also took a primary cortical neural

cultures for Mouse and we grew them as a comparison because we wanted to make sure that we were using at least in some sense bona fide cortical neurons and the best way to do that is to take it from animals uh but fortunately as you see we will be able to move and we have moved away from that now in our current practice so we're completely uh animal testing free at the moment uh and then what we did was we plated these onto what's called a high density multi-electrode array and essentially I'll show you a bit more of this in a second this is essentially a CMOS chip which is a type of Chip that you might get in a digital camera but it's also can be used in this purpose because what it can do is sense electrical signals even very small ones and actually be able to stimulate to them and so just some evidence that we we actually did do the work we said we did here's some examples of some cultures that we've grown and they show uh what we're able to do is use techniques to actually capture key markers of aspects of these cells so for example this uh blue over here hopefully it's coming up or right on your screens this shows something called dappy which marks all the nucleus of every any cell uh if you want you can look at something called new and here new and marks neurons so all the green dots and you might be able to see a bit better in this picture or the green dots show that this is actually a neuron and then of course one of the things that we know neurons have is that they send out uh axons and that's what this thing called beta3 tubulin marks in the red and finally we want to know do they have dendrites and you can see here in the purple this is marking dendrites also a little bit of axons but it also marks dendrites as well and then more than that we want to know is not just are they neurons because on your or most neurons will have those traits but also are they cortical neurons and uh brn1 is actually a marker for cortical neuron so you can see here actually not only are they neurons but most of those neurons are cortical uh specifically there's of course going to be different cell types in there for example gfap here Mark supporting cell types such as collia and you can see here we have a number of glia you can also really obviously see that these glia these in the bigger picture have quite a different morphology than the neurons so it's nice that we have sort of this mixed population because it means that we're able to test something that has some degree of uh you know comparable uh characteristics to what you might see in the uh cortical you know in an animal or now quarter cortical uh areas uh of course it's much much simpler magnitude and magnitude simpler than that but it is you know reassuring for the function to know that there are some uh similarities as a model and importantly for anyone who has looked at sort of creating cells out of stem cells one of the big problems is that not all of those cells might be fully differentiated and if you do have cells that aren't differentiated and still polypodent when

you put them in a dish they can turn into anything and of course most of the time what they turn into something uh invasive and potentially cancerous depending what it is fortunately this ki-67 marks dividing cells so this would sort of mark any cell that wasn't uh fully differentiated and we had uh this is a case where we had none of them and that was common but generally we had none to very very few and because of that we're actually able to have these cultures growing for extended periods of time and this is what's a cryoscanning electron microscopy image so these are real biological neurons they're a little cracked because when you do the the cryo aspect of the scanning and you phrase them uh for the necessary preparation they do tend to crack a bit but you can still mostly see that there are a lot of intact neurons a lot of connections and then behind that you can actually see all the electrodes and you can also see the density and the complexity of this and this is actually an area of the chip that is less dense and and interconnected than some of the other areas because if you look at the ones that are highly dense and connected you can't see the the chip behind him we kind of wanted to show the fact that there's a chip behind this uh and I think like one of the key things to take away from this simply is the degree of complexity you get here compared to something like an artificial neural network a lot of the time people look at models and they're quite nice and neat and symmetrical and that's just not what happens in biology and as we'll discuss later there's there's probably some interesting features that arise out of this degree and this extent of uh random or pseudo-random connectivity and so what we could then do with this technology is map out this is an example of like the chip surface and then the firing rate in hertz can be seen here as a color scale and we can see over time with different cell types we can get their activity and different cell types sort of turn on or begin to become active at different times but importantly what we can do is get a fairly even distribution of active cells over the surface area fairly consistently again anyone who works in this area knows there's a fair bit of variation inherently and that stuff we're working on now for the future but we are able to get this degree of um of activity over time and that gives us some useful cell cultures to test of course uh how do we go about testing that which is the big question and uh we know that as a living organism you me mice cats whatever we need to have a closed-loop feedback to operate and what does that mean essentially it means that we're able to uh receive information about the outcomes of our actions and modify our actions as a result of it so if I reach over for a cell phone I immediately get sensory feedback in a number of modalities to as I do that action and as I pick it up I'm aware of the success of it if I fail I'm aware of that too and this is actually this little mouse with a VR headset on is from this athenshire study which is one of my

favorite studies that's been done in recent times where they actually attach it to their virtual reality to mice and they found that if they disconnected the mouse's movement as it was developing from the actual visual information it got it led to quite catastrophic uh developmental functional developmental impairments and it just really shows it shouldn't be surprising so much but it does show us like that this link with our environment is incredibly important for us as organisms uh to develop and so if you're looking at cells in a dish and you want to try and elicit intelligence you can immediately say well you need to have some sort of linkage some embodiment which is achieved through this closed loop system uh to be set up and so like one of the better ways to sort of do this is to contrast with an open feedback system because that's what people generally do so just to emphasize this closely it means that we stimulate some sort of sensation and that can be done with an electrical stimulation and we can provide then sensory feedback again through electrical stimulation on how that response Alters the world and it's actually been demonstrated before in a few studies for example and it's back in paper that this closed loop stimulation does result in increased plasticity um but we wanted to try and take it a little further and actually see if we could elicit a very clear gold directed behavior and so we set up this is a little schematic of the dish and what we did was we almost arbitrarily defined uh that there are I shouldn't say cyber Terry because there was a lot of uh planning and strategy behind it but what I mean by this is that within a given framework for example here what we did was we designated a sensory region where we put information into electrodes using a combination of rate and place coding and then we read it out in a sort of counterbalanced region here so activity in in these two regions would make the paddle move up activity and these two regions would make the paddle move down uh but the reason I say it's arbitrary was that we were able to uh basically change this configuration to a number of ways and for the most part although there were some differences get a degree of statistically significant performance and that of course just shows the fact that neurons as again shouldn't be a surprise to anybody are incredibly plastic and able to adapt and change uh how they respond uh otherwise and I think that's the real interesting thing because of course starting out here there is no real reason why neurons should behave in any particular way unless they're doing it to a call to some fundamental imperative uh and this is just an example of the visualization you're actually able to go uh there's a there's a site if you jump to our website I'll show that later it's called Spike stream and you can go and play around with this yourself it's available so this is an example of the electrode uh multilateral Ray each Little Dot is a potential electrode we can route the blue ones are the ones we're routing at the moment each

little bar is a activity and you can see we put activity in the top and then we read it out at the bottom and it moves these uh paddle and over time as I'll show you it gets better feel free to go to the spike stream and have a look at that but right now what I want to discuss is yes it's all well and good that a system may be able to organize Its Behavior here in a changing environment and this is actually an example of a very successful so these are actual neurons driving this paddle so it's an example of one of a one of our most successful cultures but by no means is it a uh you know a weird exception we do see the sort of better performance fairly often again there is variation of course and one of the ways that we he wanted to investigate this initially was this idea of the free energy principle which I'm sure many of you are familiar with so I'm not going to go into it in a great amount of detail but I will provide kind of like a higher level summary of it here and the key things that we can sort of boil it down to in in a very oversimplified sense essentially it means that what we want to do with the frenzy principle it suggests that there is an innate imperative or a system to actually match a prediction that it makes to incoming sensation and it's going to be phrased in Bayesian terms and there's a lot of implications one can take from that and you know I love this picture that um carlston wrote up in his uh or developed as part of his Nature Neuroscience reviews uh article if you haven't seen Reddit it's a great article it covers a lot of good stuff but simply between the linkages between the different levels of the system and of course like it is in like one of the strong Arguments for it is that being able to predict and act in your environment is theoretically necessary so and the other like implication of course is that there needs to be a statistical boundary between the internal and the external which is called a marker blanket and we're going to touch on that a bit more later um diving into the notion of active inference uh via the free energy principle uh formerly it's sort of been stated that this is a process of minimizing variational free energy through action and perception uh and essentially what it's saying here is that a living organism is going to actively generate this internal model of the external world and then align those models and broadly speaking there's two ways we can do that either perception or action so if I use my cell phone as an example before if I reach over pick it up uh what I've done is I've created a model in my head based on the visual information I've received uh where the cell phone is I've reached over I've picked it up very good my perception matched matched my reality let's say though that I didn't and I dropped it as I tried to pick it up well what could I do either I could get better at liquid you know predicting the world uh and say like when I reached when I reach I will always drop my cell phone or you know I could get better at defining where it is in the world all what I could do is get better at

the actual reaching action and really when you boil down to it these are the only two ways that a system can get better at predicting its environment either through getting better prediction or getting better control in the environment and so for our cell cultures what we wanted to do was simply remove one of these options so it's a very simple implication from a very nuanced Theory but following that we sort of said if the world became truly random something that like we as people like you can't answer most things too much because like we as people can't really experience something that's truly random um but in a dish with random random noise you could at least approximate it um and so well that's what we did and so we had these two options so what we then said is that this is true what the cells will do is actually change their actions uh to make it Accord with the they're the internal model of the world that they have to decide and derive from the sensory stimulation we feed them so under this this model here that's sort of represented in this picture that you can see what we propose is that the neural cells or the system will try to minimize the difference between the internal and the external world to minimize the free energy of the system I.E minimize that surprisal and we kind of predicted whether reversible is the right word or not uh it would certainly we'd say that non-predictable round of simulation should be saying that the cells would try to avoid while on the flip side if we gave them a predictable stimulus when they did something they liked that might be reinforcing and again whether I put these words in quotes because there's a whole language question no one could a question around what is it is a truly aversive is it driving is it whatever but for the sake of communication we have to pick words right now so hopefully you understand the uh the context in which they're used and so what we propose then is that the cells should modify their activity to avoid adopting a state in this case that's a pattern of activity that would lead to an increase in Surprise and so what we could see here we set up some experimental tests and we compared a number of different methods so we compared this control group this was simply meteor in a dish that had the stimulations and feedback applied to it media in a dish should not be able to learn without any cells we had an insilico model that would drive the pattern with random noise and essentially this was to test the control group you can think the media only control group I should say you can think of it as testing were we able to get activities simply through our stimulation the in silico was saying could we get performance simply through something in the system itself you know making sure that the system isn't biased in any obvious way then we had a rest control group where this there were cells in a dish but no stimulation and this is asking the question left alone will the spontaneous activity of neurons because neurons always have activity regardless of whether you're

stimulating them or not will that lead to better performance again there's no reason it should if our system is set up properly of course if there's a bias it would but but that's what we're testing here and then finally we had both Mouse and human cells that were given the information where the ball was were given feedback and we're able to play the game and what you can see is basically this is a little hard to interpret this graph if you don't stare at heat maps all day but if you compare say just the top right which is uh the number of hits that are occurring in any given minute by the minute on the the x-axis you can see that it doesn't really change too much over time for the control group but compared and obviously the other the in silicone the resting controls are kind of the same as that if you compare it to the mouse and say the human cultures you can see quite a clear difference here and if you want uh more more asterisks for significance because because who doesn't love asterix's uh at least all scientists love asterisks uh or other other symbols of significance what we can do is you can see over time uh this control groups uh the media controlled in silicone control the rest control they they don't show any difference over time a bit of variation But ultimately no no real significant difference over time in contrast to Mouse and the human control of human cells uh both showed learning over time and by the second time point they are performed all the other cultures of all the world cultures all the other conditions I should say uh interestingly the human cells even ended up slightly outperforming the the mouse cells and although I would be careful not to over interpret this it was interesting to note that uh We've run this experiment a number of times and this is a very consistent difference and so it could be an interesting way to start to investigate differences in the future between species uh if one wanted to set up a study more rigorously to actually test us and these results were replicated it wasn't like we found one measure that suddenly looked like performance and we ignored the rest we looked at a number of them you know some of the other interesting ones the stuff like the percentage of long rallies where it's hit the ball more than three times in a row so for four plus in a row and you can see here again significantly did you know changes over the time for the mouse and human cells nothing for the others likewise the aces which is kind of like tennis where they missed the ball without even hitting it once this decreases uh statistic statistically significantly over time for the mouse and human cortical cells not for the others so much uh so that's interesting but you know we sort of say like well it's interesting it shouldn't be surprising because we know that neural cells are adaptive uh and the fact you can see it in a dish is cool but what what are the implications what can this actually teach us about how these cells are behaving and one of the things people have looked at uh in detail previously uh is functional

plasticity so what we looked at here is of course we also assessed functional plasticity and so that there was a massive difference between the resting uh State functional plasticity and the gameplay functional plasticity but what what does that actually mean what does that actually look like going Beyond just the fact that there's plasticity occurring in the culture so what you can do is you can look here at just a correlation in the activity between the cells in the motor regions that I've shown you before and the sensory regions and it's not too surprising that whether or not they're playing the game you have quite a high degree of correlation the correlations at rest are slightly stronger than the correlations of gameplay but what is interesting actually is if you take the average correlation right not just not just what is the overall correlation but you actually break down the average correlation per culture uh and session what you do is you actually get a far higher average correlation uh between uh the sensory and motor regions uh when they're playing the game then when they're at risk this makes sense if you need to coordinate your action with incoming stimulation you need to have a degree of correlation between where you receive that action and the activity there and where you're at your activity output is going out but conversely if you have in this case basically you can think of it as two buttons even though they're spread over four regions what you would actually need to have is a negative correlation or reduce correlation between these regions uh and if you look at the percentage of exclusive motor region activity in this graph which is where you have activity in either the up or the down region and bear in mind again these are counterbalanced so this results actually pretty interesting to see uh what you see here is that during rest versus gameplay you actually have a higher degree of uh exclusive motor region activity which would of course be necessary for discrete control in One Direction versus the other and that was this was quite an exciting result because it was really supportive sort of that there is something quite Dynamic and interesting going on uh in in the way that these cultures are reorganized interactivity and likewise this actually just shows the degree that this correlation decreases over time with two linear regressions and you get two very different linear regressions over the first five minutes and the last 15 and we arbitrarily just based on some early work decided to and again not super arbitrarily but based on some early work I should say we decided to sort of split the games up into the first and stabilize and so that was quite supportive to suggest like what we're seeing here is really the tapping in of some of the key uh adaptive properties of the neural cultures and to take it further a bit more of a bit more of a focus on the sort of free energy implications here uh we can of course reframe surprise it has a link with information entropy um and you know as for those of you familiarism must be

captured as a tail just uh Divergence between the two distributions um of what might be occurring and so we wanted to actually calculate here the information entropy of the actual cultures and now it's important to note here this is internal to the cultures uh both had gameplay and at rest and if we didn't normalize it there's quite a lot of variation going on in any given culture but it becomes much clearer once you do normalize it uh simply to to baselines uh and you can see here that when you give them random feedback so one one thing I'll say firstly is that the gameplay normally actually has much lower information entropy than when they're at rest two when you give them random feedback their information entropy increases massively thereby suggesting that these cells indeed should be uh you know under the free energy principle should be predisposed to avoid this um and that that's actually what we saw but what we're interested in is to say well what if we tried different types of feedback like how do we know like this is all well and good but how do we know that this type of feedback is actually what's leading to to these results because it could be it could be anything uh and so we set out and we introduced a few other control conditions so this is our regular one that I've sort of was showing you before if it was a bit unclear hopefully this helps to clear it up a bit uh when they played the game they missed the ball they get random unpredictable sensory feedback when they hit the ball they get predictable we then added something called The Silent condition which is when they missed a ball they basically had all their feedback removed And yet when they hit the ball nothing they didn't get any additional feedback the game just continued now it is important to note of course that when the game restarts after a miss the direction of the ball is random so there is still some Randomness in this game but presumably less so than in the normal stem condition and this can be contrasted with this open loop or no feedback condition um they're sending them in this case basically where the game is played information about where the ball is is presented to the uh cells or the system cells or controls whatever the case is and then uh but if they miss they don't get anything changed if they hit they don't get anything changed and so really it's asking a question like will a culture just play the game for the sake of playing the game like there's no reason it should but maybe they like maybe they do so we wanted to test that and the final condition we included here of course is a rest condition and if this helps to make it clear if you're more graphical uh this can give you an example so here you can see the ball going up it hits to it and then you get like a predictable stimulation here the ball goes it continues and it gets hit again and then it misses and it gets a random feedback and then the game continues silent in contrast you can see that this random feedback here at the same time point is removed for the silent condition likewise

there's no predictable stimulation across the culture and then the no feedback it's just the ball becomes completely predictable from the very first serve there is no unpredictability in it at all uh and so when we look at the results here and obviously if you're interested you can go look at the paper because we dive into this in in you know similar detail as before but uh just in brief what I'll show you is that here what we've done is just compared it to rest because it helps control the variability a little bit um and it makes it look a little neater uh but essentially the stimulus condition they outperform rest they outperform their starting position over time it replicates what we saw in the first study and The Silent One there is a very slight increase it doesn't reach statistical significance but they do end up also statistically slightly better than the no feedback condition which shows pretty much no learning compared to rest at all so that matches up pretty much with what we expect to see like around the free energy principle you know a condition that gets a small amount of Randomness shows a small amount of learning if you want to do that but not a lot uh something that gets a lot of random information as a result of an action that we deem undesirable shows a greater tendency to avoid that action and something that has no incentive to behave in a particular way doesn't behave in a particular way so what was really interesting and exciting for us also although I'll be honest it made us stop for a few minutes to actually think about why this has happened was that when you look at this normalized information entropy again we got a replication that when you gave them random feedback the the uh actual internal information entropy goes up with the silent condition the information entropy also went up and in fact even more so and we went why would this be the case and of course what we realized quite quickly was well neurons left alone do fire spontaneously and when you take a neuron from a stimulated condition to an unstimulated condition they do tend to re-engage with the random spontaneous activity quite vigorously straight away and so we went and yet it didn't show the same degree of learning so of course like I will say this is one finding that absolutely needs to be followed up more in the future and we are doing that but it does have an interesting implication in terms of Markov blankets so I mentioned this before I flagged it before and this is why because I think this is one of the really interesting results uh so to go into more detail what a marker blanket is is it's a statistical boundary that can distinguish an internal state from an external State and this can be at any level so it could be at an individual Neuron a cluster of neurons um and nuclear distinct functional region of the brain the whole brain the body in relation to the rest of the world or what's outside it and so the fact that as I was saying that we see that the silent condition even shows more random random uh or Randomness or information entropy then

when we're actually providing external random information into the system uh is actually very very consistent with what we know about the stochasticity of spontaneous activity but what it does also suggest is that cells will respond differently to Randomness inside a system versus outside a system and that was a really exciting result to sort of find like almost physical evidence or like perhaps you know this marker blanket like really exists of course it it makes sense it exists like theoretically there's a lot of reasons I mean even just the simplest notion that we can separate what our internal thoughts are from someone's external thoughts uh it's so not it's not external thought someone's external voice to us I should say uh suggests that we need to have some sort of barrier to distinguish these types of uh structured information one internally One external uh but seeing it was very very interesting uh and so we also wanted to of course extend beyond this and sort of not just stop here and look at what other interesting things might be happening and you know one of the other interesting findings that we've we've recently completed and there's a pre-print of this up now is this idea of neural criticality and so what is criticality for those who aren't very familiar with it in brief it is this state where a population's activity is coordinated uh basically between two two other states so you know physically you could think of it at that point where between where water becomes ice and there's a point just in between which is a transition to bring it back to the brain you can think of it as a brain connect firing rate could be either highly coordinated rhythmic oscillatory much as like when we sleep or in Catatonia or it could be completely disorganized like in some cases of epilepsy and that then there's a middle ground and there's obviously a quite a you know it's a spectrum so there's quite a lot of middle ground but the closer you get to this sort of critical transition point this is called criticality and theoretically it's been proposed to maximize uh give me there's my timer telling me to take a break it's theoretically it's been proposed to act as a key criteria to maximize information transmission and capacity and what's also really interesting is that it's been linked to a number of different cognitive behaviors although the field is certainly a bit unclear about what that is so it's been like for example two responses to drugs working memory attention one of my particular favorites is cause of fluid intelligence uh but these Studies have some some concerns with how they investigate it and ultimately one of the biggest concerns is that they're doing these you know these these ones at least in uh humans we have lots of compensatory mechanisms you have to use uh fmri or similar mostly fmri measures to actually investigate it so there's a lot of barriers in actually concluding this so we set up a study to actually investigate this and essentially again to not go into too much detail about the nuances we looked at three key different metrics there's been a

bit of criticism in the field where people focus predominantly on power laws power laws can arise from noise alone so we wanted to avoid that shortfall and look at three different measures and so this is this is sort of the setup we had we took them we compared them either playing the game or at rest which I described to you before and then we looked at these things called the the DCC the branching Ratio or the shape collapse error and we wanted to see how we could group them my slides have Frozen let me try to jump to the next one alrighty so hopefully that's back for everyone now let me just there we go okay so what we saw here really surprisingly was a very Stark difference in in criticality uh across a number of measures so you can just see here basically the prediction between the the rest and the active was it's quite a just noticeably by eye very different and of course that statistically significant diving into a bit more detail we could look at these different measures and this is just some quick overview of what those different measures look like um so the the DCC the branching ratio so for example with branching ratio critical systems sort of maintain the number of uh uh sort of threads of activity that might happen in a subcritical system that variation will die off over time in a supercritical system it's going to continue to expand up into a point and then there's a shape collapse error of course which is looking to see how you can collapse different Avalanche shapes across a given spectrum and then looking across all of these we get quite strong long statistical differences showing quite clearly that when you have the system embodied in an environment it's showing this measure of criticality which is very interesting um and one of the interesting things for this actually is that it shows that this criticality is very fundamental and that information input seems to to drive it very strongly and to us this does seem like a challenge suggesting that it is in itself inherently a marker of some sort of higher order system but of course I think the most interesting thing is simply that we can identify this and then what it does is open up the possibility to further investigate it with more nuanced tests that were explicitly designed to investigate criticality but what was nice also to see like I guess some suggestion that it does have a role in some sort of information processing in some degree is the fact that he is also just showing you these results in a slightly different measure but looking at the correlations there were some artistically significant correlations between the different measures of criticality and the gameplay performance showing that cultures that showed higher degrees of criticality across all three markers also had better gameplay performance in this hitness ratio where they hit the ball more often than they missed it which was of course interesting to see and does suggest that although we think that because this was arising in such a simple system it uh it was such simple input that it's unlikely to represent or

at least in of itself be a criteria for intelligence per se that there certainly is some link between processing information and criticality and if you want a more nuanced discussion of this um please feel free there's a there's a pre-print up now while the papers going through clear review we discuss it in more detail there and finally what I would say though is that just to sort of really Hammer home the point that this is a very fundamental property of these cultures and that it is so drastically different when they're playing the game versus rest that when we wanted to classify the cells based on criticality metrics and performance as either playing the game will rest we were able to do this with a 98 success rate uh if you took into performance and if you even if you just ignored performance and you just looked at criticality with random Forest we're able to do this with a 92 percent success rate of characterizing whether they're playing the game or at rest and that's that's obviously suggests that this is very fundamental that one says information input in these cells uh they drastically reorganize their activity and that's really interesting of course one of the questions that we also get asked a lot when we sort of put out this early work was a lot of people would say ah but who cares it you know we literally had someone write a comment one of the peer reviewers not in the uh actually at one of the points said to us you know but you can't this can't actually do anything better than reinforcement learning so how can you conclude anything about biology which was a surprising comment to get um but what what you know we wanted to do and said fair enough like you know we think there's plenty you can learn about biology whether or not it does better or worsen reinforcement learning but it's a fair question like does this do you know performance wise can it do anything interesting compared to reinforcement learning we know that if you run reinforcement learning enough you will get superhuman performance with the game of pong they will solve it they'll beat humans that's fine they can do the same with protein foldings chess go that's that's all well and good but that does that mean that there is no room for a biological system uh when it comes to this so what we wanted to do was actually compare it for sample efficiency because that is of course something that reinforcement learning uh struggles with greatly so what we did was we compared three different deep learning algorithms dqn a2c and PPO with three different types of information input and against our biological cultures to see how they performed and so just showing you here some results where we took the image Vector which had a CNN to run and process it and then feed it into the the dfri algorithms what we're able to see actually um was that a mouse in human cells which are here in blue and orange significantly outperformed and out Learned dqn was was pretty much always the lowest performer which is fine we know that it's very sample and

efficient but even more sample efficient algorithms like a to c and PPO also failed to show the same degree of performance even though they started higher for whatever reason we could never quite figure out to some Quirk of the algorithm they never showed the same degree of learning and this was again mimicked across multiple comparisons but what was brought up to us was this idea of saying well there is of course the character dimensionality so what you're doing is you're feeding in a I think it was a 40 by 40 image Vector uh to these these the Deep reinforcement learning uh and of course it's going to take them longer to converge they have to deal with more information and we went you know what that's a fair point so what we did was run it even simpler and we came out and we said all right what we'll do is we'll feed them vectors of Simply you know four Vector uh Matrix of where the paddle is and where the ball is and that will be much simpler and closer to the game plane so we fed that into them and what we saw was actually they performed us with less information uh and of course these are the the mouse and human image time because we can't alter the information put in this way it was a post-hoc party the actual Mouse and human cortical cells but yeah they did even worse across multiple matter and we found again they're outperforming it a little more information than what the uh also getting because their sales are really only just received in the ball position we don't we're not yet able to encode proprioception because the disk brain system is very simple and you know although we would argue again like this sparsity of information should surely make it harder not easier with less information and certainly with our initial pilot testing we found that the more information you gave them the better they could do we again said fair enough let's change the change the information and put to the uh reinforcement learning algorithms give them the simplest possible information that matches as close as we could possibly do to what the cells are getting and see how they performed and again they performed even worse suggesting that in fact more information is probably helpful uh and certainly what we could show here is that like yes there is certainly something interesting going on in neural cells and the way they're able to process information and it comes back to this quote that I love so much which is to say that like if the human brain or in this case even a simple collection of neurons in a dish was so simple that we could understand it we might be so simple that we couldn't uh and you know we like to think that the tools we're building such as this is going to help us to actually simplify it enough that we'll be able to unpick and start to understand the nuances behind this but one of the nice things that we're able to show is that even without fully understanding all the nuances of what's going on inside a dish like this we can see that it does have something strange distinct traits in terms of intelligence that have

promising advantages for information processing above and beyond what's going on in reinforcement learning and we really think this happens home that this is going to be a powerful tool to investigate these features in more detail so as a as sort of a summed up conclusion uh what we're able to do with this system essentially is have a system of real biological neural cultures that can exhibit a very rudimentary form of uh natural intelligence uh through their inherent adaptive traits and these adaptive traits uh like frankly quite amazing to the extent and the variety and diversity of what they're shown and even with the data that we've really generated we're only just scratching the surface but there's a lot of potential that they have and so simply by having this closed-loop system of electrophysiological stimulation and recording via these multi-electrode arrays we could embed them into a or embody them into a simulated World depending which term you prefer and they could get better at playing it and moving the paddle to actually be able to perform the game uh with quite a quite a consistent degree of coordination with a faster learning rate like then multiple reinforcement learning algorithms and uh you know of course this is as I said it's a lot of work we're only just scratching the surface and so one of the things we're excited to do because we're we're actually not an academic lab we're a small startup based in Melbourne Australia because we want to open this platform up to everybody so if you have a question or you think everything I've just told you is completely wrong and that neurons work in a totally different way that's fine you know reach out we will in next year um be opening up for Alpha Testing and so the most exciting projects we'll be supporting to have an early access to this if you want to investigate the properties of these neurons and you have basic python or Java coding skills you can access them virtually and code them as an environment and see how they respond and test it and you know go away and write a paper sort of showing how how you think that it should all work based on actual data and this is because we acknowledge this is a whole whole industry like this or whole field of research of course and so we want to try and make these tools available for as many people as possible and we're going to make it as easy for you as possible as well and it's going to be fairly affordable so that's that's the end of the talk and uh of course it's not just me um working on it there's a whole team and a number of collaborators that we work with these are just the collaborators that sort of touch on this work there are many other very valuable people we work with that are expanding in other areas and so and actually we're fortunate enough that uh Professor Adele razi who who's one of our closest collaborators has actually been able to join us today for the discussion part of it uh and so if there are any discussions really happy to to take questions all right wow well uh thank you very much for

that excellent presentation it gives us a lot to jump into so perhaps a deal feel free to say hello introduce yourself and add any general remarks so that we can jump into some more questions yeah thank you Daniel great talk right and as always you did uh it was intuitive fools that there is so many projects that you love is uh is really doing amazing work so um I'm based at Menasha University we're not that far from the um cortical Labs is um yeah in the city really but it's a Suburban at the main campus University where we have this computational systems near science lab and our lab is using brain Imaging and computational models uh especially Dynamic causal modeling and active infants as Frameworks to look into the brain and try to understand what's up you know um yeah how one can look at little mechanisms of for example how the brain implements coordination for example uh for this we think we we use as I said um mathematical framework of divine friends and also dynamical modeling so that's basically what we are doing um and happy to contribute to this discussion it spreads um you know um you know there and it should be uh the one who should be talking more than I would be gladly just sitting here and listening to wonderful uh interaction we already are having awesome well my first question just for context was how did active inference come together with this line of research were you studying active inference and then looking for an embodied system or was it something like you were studying this very interesting system and looking for some type of framework to help you model absolutely it's a great question and it goes back it goes back historically so um Andy kitchen actually who and uh and and han han who's uh one of one of the founders of the company uh they were they were interested they were looking for for ways to explore it and actually a deal uh you you ended up having a conversation with them I believe and they were basically looking for Immaculate of like how could we how could we talk to the cells and they came across active inference uh or did you introduce them to active inference um I would say that that Andy and Han at that time that we are talking about late 2018 early 2019 um and they were already interested in freeing the Japan support um and but then I just moved to Melbourne from from London in mid-2018 and I ended up giving a talk at Alfred where cortical lab is a big incubated at the time um and I was discussing these uh these ideas and it was kind of uh you know um interaction that just happened and then we thought yeah well okay there's lots of synergy and there we go uh we had uh we workshopped some of the early ideas uh at that time um and thought about you know on and Andy were very thinking about how to actually what could be the first thing that we should do which would then um really uh showcase the power of the discipline of course the discipline exact name came later on yeah yeah that probably has been sampling child no no actually this

Frame describe is a name definitely goes goes to Andy as well um okay I would have given it some some sort of uh esoteric uh name that everyone would have gone what the hell is that so no it's a good call by Andy to do that um yeah yeah but of course like it does have a lot of really useful features uh the whole active inference framework uh that that make it quite an ideal thing to test you know the whole action perception cycle the fact that there are like implications that you can simply take and be able to test uh the biological plausibility of it uh or make it very useful as a system for us us to adopt um having said that though and like a you know deal can attest to this like we also well I think that it was uh I think as a Neo Seth calls it uh maybe other people call it that too it was a bit of an adversarial uh collaboration in some aspects as well because we were interested in the idea but we had no stake in proving or disproving it we were simply testing it and so we had a number of discussions as we went through you know what if we see X Y and Z uh what would be the states just kind of challenge it not just support it and so I think that's one of the things that made it a really exciting collaboration in in that there was a high degree of skepticism amongst the team about pretty much everything we did um and I think that's what came together and making quite a nice paper because we were like as everyone should be in science but let's be realistic seldom are we were trying to be our own worst critics because we were trying to figure out like what else could be explaining it and you know and I think there are interesting future directions we can explore but certainly the results as they are did find support that this only is a key characteristic of how cells respond awesome so one preemptive question about learning and then we'll drive back towards active inference so what kinds of mechanisms do the neurons use to engage game mode and also to learn are they changing their topology in terms of their connectivity are they changing what happens at synapses are there changes happening inside of cells what happens when the game mode turns on and how does it actually improve throughout the course of that experiment uh look that's a great question and the the reality is we don't have the full answer for that yet um most likely it's going to be a combination of the latitude you mentioned changes that are happening potentially inside cells or inside synapses I mean it's not necessarily too useful to try and break those apart the idea of like larger scale morphological changes are less likely simply due to the time scale that we see and the fact that we don't typically um although there will be some future work sort of showing some other things at the you know as we've moved on and Advance with our work and we typically in this study at least didn't see uh learning across days so it's unlikely there are massive changes in sort of connectivity it's more likely that it's functional changes in plasticity and how they

respond to information and it could be a number of things like it could be um you know of course there's probably some sort of relationship and balance between heavy and homeostatic plasticity long-term potentiation to long-term depression and then you there's so many mechanisms that you can break down to look at how that is uh you know from you know like one interesting candidate could be looking at like phosphorylation which interestingly takes I think about five minutes to to translocate from from inside to outside uh synapses and then is known to alter long-term potentiation which would roughly Accord with that time but this is something we need to investigate more in the future um higher level to that I mean a deal might have some some ideas to sort of discuss higher level uh than this you know less reductive and more high level yeah more more computational so um yeah I mean you know we are really interested in learning mechanisms uh and especially uh formative and transformative um what uh the learning mechanisms in artificial uh that we can mimic inactive field neural networks that so so one of the you know while we have the biological chip which can learn can we actually now understand how it's doing this and can we in the short term you know uh have in a better artificial neural networks of course the you know we have wrapped in this team is taking to us that we actually don't need that artificial neural networks anymore uh we'll just have the biological chips right um so so that's that's exciting um so we have been looking at you know certain things some of the more um unresolved problems in in deep learning and in machine learning for example um lifelong learning and continue learning uh where we know that artificial neural networks they break down in in in in tasks where you have to do multiple tasks Industries and they could do very good uh human-like performances on a single year task but then uh you train them on the second one and then they forget what they learned in the first one unlike humans so it could run swim speak without you know forgetting what um you know otherwise it will be a catastrophic um so that's one thing another thing um you know credit assignment so one of the bigger problems how do you actually uh solve the credit retirement problem which is again an also a long-standing problem in deep learning that if so what sort of uh choices uh we make that result in uh rewards and future so what uh and and and many a times what happens there's a delay there's a delay in recording what behaviors resulted in the uh in the in in the reward for example uh so so looking at what elements and what choices that we made that resulted in a long-term gain is another problem that that I I we really don't know right now how do we actually going to but these are bigger very big problems which are unsolved and I think it just shows you the the power of the dish braid system that it can actually help us to you know I think there's a real um real opportunity

here for us to actually attack those problems as well at the you know how level yeah as Brett was saying and I think it's it comes down to also being like the right tool for the right job so reinforcement learning cnns can do amazing things uh if you've got the data to train them if you've got you know the uh the rest of the setup that you need and you want to use it for the right purpose are they very good for doing stuff in Dynamic Real Time with limited data fuzzy information but so far no um and so maybe like there will be a case where we can take some of the findings from how neurons work and apply it in an algorithm in itself uh you know I know a name a deal and and his his team are doing some amazing work with uh you know building up active imprints models uh and I think those results are really going to you know I think excite a lot of people just as just as just as some flagging of some of the work you're doing a deal I won't mention anything else but it's going to be really exciting I think for people to see what's what's happening um and but likewise I do think that there will be things that biology maintains superiority over uh the question and the nice thing about biology of course when you talk about generalized intelligence is we have no proof of principle that generalized intelligence can arise out of silicon Hardware or Quantum hard you know Quantum based Hardware we do have proof of principle that it can Arise at a biological wet wear our brains and brains are flies of cats of rats whatever uh show general intelligence to differing degrees and so to us it's like not a question of can biology show generalized intelligence but how do you get there and that's a less speculative question not necessarily an easier question to answer it's not easier necessarily but it does give you that ground truth it's the thing in and of itself sort of answer wow so much there in silico and in principle systems can display open-ended symbolic learning going back to the turing machine but for the kind of open-ended enacted intelligence that we might be interested in in settings like you described which are real time fuzzy and with small data and we could also add with a high risk or survival threat those kinds of settings we have empirical evidence across the surface of the planet that biological systems can make that happen and we have zero empirical examples of silicon chips making that kind of a functionality arise so that's a very interesting point that kind of brings to this question about the similarities and the differences between the artificial neural networks and the biological structure and also between the reinforcement learning Paradigm and the active inference Paradigm in which rather than maximizing a reward function we are minimizing an expected free energy or a variational free energy by way of bounding surprise so in describing the experimental setup the reward or rather the consequence of succeeding or failing was you adjusting the regularity of the input and in a reinforcement learning Paradigm that's a little bit

of an unconventional manipulation one might expect oh we'll we'll add sugar or we'll add a dopamine Agonist or an antagonist to the dish to Signal the positive or the negative valence directly kind of by putting our finger on the scale of the reward function but rather you were able to train and Learn by changing the patterns of regularity of feedback so how does active inference have similarity and difference with reinforcement learning in terms of the way that we can use patterns of regularity not just valence of interventions in learning I'll give an answer that I'm going to pass over to a deal who who can dive a lot more into the computational side of it than than what I what I could um what I would say though is that there's a few things as well like one of them of course is uh can you even adopt a reinforcement learning Paradigm in a dish uh it's not like there was a choice between the two because we don't have access to a privileged reward Network inside the dish to be able to say you know good sales bad sales we you CL you could like in theory Square dopamine in but you can't continually Square dopamine and you can't do it in a spatial or temporal resolution that that's accurate enough right it diffuses slowly across media we can't do it um so for us we needed to find something that was more fundamental and I think that's an important point and I think it's also interesting to note like that there are studies that look at the fact that you can mimic changes that you see with uh electrical information that you can also achieve through like the application of small molecules such as such as dopamine or whatever neurotransmitters um and the right stimulation can increase long-term potentiation much in the same way that say dopamine would and so dopamine seems likely to be a tool this is speculative of course but it seems likely to be a tool that shortcuts uh the the way that these systems work in line with some broader imperatives such as free energy minimization more so than uh being something unique and special it is unique and special in its own right like in the way that it works but it must be serving a more fundamental function um yeah I think that that's probably like a good way then to sort of lead into the differences between let's say an active frame inference framework and a reinforcement learning framework which I think a deal would be best to best to discuss yeah yeah um great question Daniel uh first you started with the differences in uh in silico models uh artificial neural networks and the biological neural metrics so uh first of all as we see uh the artificial neural network is a um is a caricature of an actual biological layer on um it just mimics that you know it takes an input uh you know and then gives you an output there's a non-linear you know function and then once you have many of um you know and stack them up and and make lists they can do amazing things so this is like new artificial neural networks just uh using the one of the video uh a cartoon of

what a biological neural network do it would do it it has a lot more than that it has it is an extremely complex uh biological uh you know things so uh imagine now that uh you know we can reuse all of the Power and all the uh you know you know um when it comes to what it can in terms of information processing and then uh build systems how powerful they will be um as we know you know uh currently these huge um what we call uh what we call these large scale language models for example llms and and then all those fundamental models that we have been you know seeing uh with billions of parameters to train them it requires a gigawatts of power it's basically like a whole city can be looked up with once they train not a human brain is just using a you know a small fraction of in area time and power to do much more than what they could do they still can't do so causal learning causal reasoning problems very basics of them uh involve that logical neurons can do it you know ways whereas as we are doing it right now we are having a back and forth discussion and you know um that a big is one of the most uh sophisticated sophisticated and he age neural network would not be able to achieve uh even for a few few minutes um so that's that's what we are looking at there's an enormous um you know uh opportunity that we have to actually harness this power and do amazing things um now coming back to so that's you know uh now coming back to active inference and reinforcement learning now this has been the uh this has been asked many times and um there are papers that colors it in nothing new side it has a paper on on where they look at um you know however um pnm what is Bala so I think active inference is biologically plausible so uh so the main difference is is that when you are uh an active infant system is is optimizing a single quantity it's what we call the free energy of Asian free energy and that gives you um that gives you with a single optimization function you are doing perception planning and decision making in the same with the same quantity so um let me put this uh this in this way uh you know things that how things have happened over the past few decades that we it has been always been about how to optimize a function so once you have let's say a value function a reward yeah uh in reinforcement learning is all about how best you can optimize this quantity while what we are asking we are saying that what is we are optimizing not how we optimize so what is that quantity and that's the relation for energy that we're optimizing and it gives you all of that in a single you know compared to a the enforcement Learning System that would require multiple objective functions to optimize so I think it's not biologically attainable to have that so I don't think uh and that's why the active inference um is a is a unifying framework and it probably would Rattle and offend free um that you know that we think that you know that active infants subsumes um all those uh you know um computational

Frameworks like reinforcement learning um you know uh into a dynamic programming and all those uh derivatives uh because it is basically what the brains are doing you know and and they're doing it better than what uh you know uh framework that we have been using so far uh so I think there are differences clear differences uh and uh however having said that I would really would like to to see uh in future and probably we probably will have a dig at it how you can actually mathematically which is then become unambiguously show um provide a proof that where these uh you know why active Inference so that there's a still this is still a gap there while we could um develop uh insular active insurance agents and we can show them that they are doing better than reinforcement learning agents and then we have quite a few um there's a whole uh sort of work that uh of my PhD student Ashwin Paul is doing and the papers are coming where we are comparing active influence agents with um with uh with the state of the art means learning and learning you know um networks um and we are showing of course the the active inferences is is is is doing better uh but the theoretical proofs are still missing similarly like how active inferences is is what are the similarities and differences for example I would be very interested to know how it's different from uh to durov's work and again Ashwin is looking at that I am an alternative paper VNS paper in 2009 which looks at efficient computations optimal action which has far reaching it they show that you can um both in discrete and on continuous settings um you know they they have shown that they algorithms can be dynamic programming and in subject learning based uh based algorithms uh what is this uh what is how active inference is different from Tish price go to information bottleneck uh you know um thing how how what are the links between Universal pride and suleimanov all those all those things guys too you know it is it's it's really early days uh you know uh what are the similarities the differences of of activations with all those big Ideas which are therefore you know in cybernetics and another literature funny many many many years uh for for decades what are the differences and in similarities so it's a good question we don't have all the answers but um you know all the yeah so so it would be very exciting next week is uh looking at all those big questions I just wanted to Echo one point you made and then ask a question from the chat so in reinforcement learning what you should optimize is almost treated as an obvious question you want to win the game you want more points that's the way to win you want to stay alive then you want to optimize survival so that's your reward function and what to optimize is kind of swept under the rug and then there's this big question of how to optimize it and that's what has led to this enormous diversification of different approaches and heuristics for how to optimize and different gradient descent methods and so on in in contrast

in active inference what to optimize is actually known it's the variational or the expected free energy as exactly specified by the generative model and we know how to optimize it which is with a variety of locally plausible biologically inspired update rules for example message passing and belief propagation so that's very subtle but also paradigmatic difference which is that the structure of what we're going to optimize as our unified imperative for perception cognition and action the free energy functional has methods and software packages that will step by step convergently optimize it and so the onus is actually to specify the generative model in a useful way rather than treating what should be optimized as just a bygone discussion and then go into all these questions about implementation which are the parts that actually differ the most across systems so I think that's very insightful I wanted to ask a question from the chat from Michael what would be the difference if you change the electrode array to stimulate differently for example instead of just growing the neurons on top what if the neuron synapsed with the electrodes so what kinds of experimental setups are possible and which ones are interesting and how do you know and what do you choose to do I think the first thing to say about that good question and certainly as the technology develops we'll be able to ask these questions in better and more interesting ways like it's probably worth well you know I've made this comparison before but I'll make it again like the work that we've done here is you know we we're excited by it but at the same time like it is probably equivalent to like the early transistors made by Shockley you know it's janky it's kind of ugly it does a job but not without not without limitations um and so you know what we're really excited in is as the community hopefully come together and work on this 5 10 15 20 years maybe longer um you know 70 years of continuous development have led from a transition to being a large ugly thing to thousands of them you know within every device that I can see uh so that's like one thing I just say in terms of like I will point out like that the the neurons do in as much as they probably ever can integrate with the electrodes as is as you saw from that picture I showed at the start they are deeply integrated into the the electrode like we cannot like we could not and let the neuron survive remove them in any meaningful way and keep that structure so you got they would have to completely like this is so share their structure and most neurons won't survive that um for very long anyway uh so they are currently integrated like I mean synapse so synapses work slightly differently uh yeah there's a whole there's a whole collection of traits like there is of course electrical activity is a shared language between neurons and and uh the hardware and that's what makes it possible to stimulate them with electricity because it can it can sort of mimic the change in ions across the the membrane to

stimulate Action potentials which is how they then communicate with electricity but an action potential is of course trigger the release of chemical signals and so forth and so there's a lot that's going on there and I think it's it's about building the bridge it's not necessarily about [Music] rather than quite the right word phrases about the completely recapturing everything that's unique to biology and the hardware all unique to the hardware in biology it's about where's the useful bridge and synthesis uh between them to elicit The Best of Both Worlds and you know again it comes down that right tool for the right job but in this case it's like a different circumstance like what's the right tool process the information uh is it is it a biological part is it the hardware part and so when we communicating then what we need to figure out and again this is an open question like I'm not sure where it will go but what we need to figure out is like where what's the the best way at the time of the technology we have right now it's it's not trying to replicate uh sort of hardware-based synapses to do sort of chemical communication although maybe in the future it will be but yeah it's a very interesting question and like the basically answer is like we don't really know where it could go and we don't know where it would do um as I mentioned in the talk like they are incredibly plastic neurons we can change the configuration we can change the type of neurons we can move from 2D to 3D and this is work that we have ongoing now um and I think that what we're going to see is just an explosion of results as as we start to make inroads in this area saying that the neurons might not survive the separation from their cyber physical digital substrate reminds me of the reality of some to all of us today if we were also stripped from our digital substrate many critical systems also would not function and I think that speaks to some of the recent work of friston at all at verses with the idea of the ecosystems of shared intelligence where we're thinking about distributed cognitive processes that are mediated by interfaces and so some emitted data set is received by another cognitive entity on the other side of the blanket and maybe it's person to person maybe it's computer to computer or something mixed or some augmented system but there's a really tractable and elegant way to talk about those heterogeneous systems so even if in the local context one algorithm or another out performs another there's going to be an immense value for using active inference and fep to just talk about how to compose these systems and so that's very important insight 100 that's a really good point Daniel like very valuable and I yeah it comes down I think like fully agree with you it comes down to like how do we phrase and think about these even and that's a challenge like even discussing what these systems are and what they're showing is a challenge right now and we need to work on building up that language together so we can have those useful

discussions and figure out how to leverage this distributed Network that we're all I think it's inevitable as you know that you bring up that that is where that's where we are and it's we're only going to go deeper down that direction awesome just a little bit of some questions on information as we head towards the end it was a very striking finding about the changes in information Dynamics when gameplay was activated versus rest as well as during learning and also that's kind of analogized by let's say a person in an fmri and they're mind-wandering they're engaging the default mode and then some task comes on and the information Dynamics sharpen and we associate that with the mechanisms of attention so how did you measure or quantify the information or what do those informational measures mean in the context of neural firing patterns um yeah that's a big question and it's again it's something that that we need to do more work at diving into to actually unpick about what it actually means at this point what we're limited to look well that's not entirely true we're not learning what we did look at just first the sake of time and complexity were individual spikes in a given location and so you can arrange that spatially and temporarily and so what we've looked at and we've dived into this more work with our recent work with organoids as well trying to pick apart sort of the sub functional subunits of computation or intelligence or whatever word you want to use um but yeah so essentially what it is is it's the pattern of activity you're seeing across time and space for a given spike in a location versus the other ones and so there's a number of ways you can look at this I think the really exciting stuff that's coming out is around sort of trying to break apart so people started hydrogen husky models Etc they looked at neural spiking and they were like how does a single neuron Spike what makes a single neuron Spike and they came up with some rules that are mostly true but now we know are kind of still wrong because you know that many neurons follow those rules and many do not and some are circumstance dependent um and then people sort of extended that and they looked at how the given uh system changed its activity over time and then of course you can expand that out and look at like how do pairs of neurons and then trios of neurons and grow that up into a whole population and so what we've tried to do with this work that I've sort of shared with you today is look at that at a few different levels and so depending the level depends the the exact answer to like what is information but um essentially like the simplest answer is it's the spikes that we can see in this case going on and their relationships to others spikes I don't know deal might have more to to expand on that um no I think you have clearly you know I think uh at some point we've also you know as you mentioned we want to look at those functional subunits considering as the as they as you know using up those main field models using uh those

functional units um as little populations and then you know uh hopefully they emerges like in a few functional subunits and then can we look at the connectivity patterns using for something like enable mass models you know which have been used quite a lot um something like a Jensen with models but a bit more uh Dynamics in them um and uh yes that would be very interesting to see uh how the patterns are emerging when neurons learn how their patterns are fine patterns and and connections and and then you know how these are changing over time as more and more learning is happening and that would be something you know we are all talking about lots of uh unknowns and lots of very interesting directions that one can take um which makes it all exciting and I think like our thinking also at least my thinking has changed over the course of looking at this when I started out I I kind of thought well an individual neuron is the smallest functional unit theoretically right so that's probably what's going on and then they'll connect up and then I realized well I think like yeah it is a single neuron it's a functional unit but so are two neurons and three neurons and and the collective you know uh you know exponential growth as you go out to whole pattern at the end of the day when we look at actually at the levels we're looking at for example criticality if you if you go to the preprint we broke it down into areas like sensory versus motor areas and we found that they all kind of behave more or less the same so does that mean that on the dish we have one functional unit or eight hundred thousand or everything in between and looking at looking at the organoid stuff that would move to that that we're not quite ready to share yet but like I think what we can sort of start to tentatively conclude from this work is that it's it is indeed that answer of like one to the combination of the the everything and I think that's what makes the biological systems and this is why I emphasize it in my talk look at the amount of connectivity and sort of chaotic connectivity to some extent um that's occurring Within These systems and that gives you something that we can't model with neuromorphic or with algorithms with any degree of uh with the same traits at least I think I can say without being too controversial with the same traits we can't yet model that algorithmically yet and I think that makes this interesting as well awesome one just remark on that before kind of a closing discussion the Hodgkin Huxley models were on large single isolated neurons and so a theme I heard you say multiple times was like the right tool or the right model in the right context and so when studying sodium and potassium flows across the membrane of a single isolated neuron that is absolutely an appropriate model and something that provided in immense insights and then as you pointed out especially if you're interested in properties of systems and connected neurons it's like a yes and like R sodium flows but you could imagine contexts

where learning happens and the informational pattern or the entropy of a given neuron does change or it might not change but the pairwise relationship might change or the pairwise relationship might not change but some higher order relationship does change and so that doesn't replace mechanistic or smaller subunit based models but rather we can broaden the discussion to thinking even about the environment like of the dish or of our peripersonal space that these cognitive systems are embedded in and talk about distributed cognitive functions without privileging the ecological or the sort of atomic levels of explanation yeah absolutely absolutely and thank you for clarifying that if I sort of implied otherwise yeah awesome no no just to highlight because they're uh despite the ring on the cover of the active inference textbook there isn't a single model to rule them all and I think like it's really by getting into the particulars of experiments and also by speaking with researchers and by talking about this kind of Cutting Edge research that we see how pluralism in the methodological and in the explanations actually plays out not just very exciting um so I guess just a uh closing question and then we can just have any other remarks uh Michael asks in the chat what should someone study or do to get more involved with this research I'm an undergraduate student interested in pursuing this field for my career and I'd love to work with cortical labs um so I think it really depends the area you want to go down like we're always looking for good people to work with and as I said before like this is going to be a whole whole industry uh and a whole field of research we we hope and believe anyway um so I guess you should pick what it is that interests you in what you're passionate about it's there's so much stuff going on right you could study electrical engineer you could study Physics you could study software engineering you could study Neuroscience you can study stem cell biology um anything in between uh these are a multi-disciplinary highly collaborative outputs um you know I showed the team before we have people from all of those Industries and and others and it's only the combination of this these these areas that uh give rise to the chance to actually make something that it's the Gestalt right it's greater than the sum of its parts and I think honestly that's one of the reasons why we've had some success coming from a sort of startup industry focus is that that's not that many of us and we don't have that much money but what we do have is like an amazing team of people to work with amazing collaborators like a deal to work with um and we can bring it all together in one spot and that's that's hard to do by its nature and Academia like by its nature Academia tends to Silo and you have exceptions to that rule of course you have like very highly collaborative works that do come out of Academia but they're always they are always I've worked in Academia um side and they're always at a distance those collaborations they're

not all Under One Roof with one one goal and one purpose this is something though Oh no just so yeah please any general comments feel free to respond to that and also just any general thoughts you want to add I was I was I was just saying I cannot recommend enough cortical Labs uh they are highly disciplinary multi-disciplinary team with expertise from cell biology to you know to software engineering to real-time interfacing to computational models and and machine learning so uh you know anything that you know your expertise and you want to do there would be something that that that's there and really cutting edge and so um yeah I think there's this whole project you know it just shows you know uh the whole breadth of things that we can do from uh from neurons to sentient behavior and I'm I'm petting my bags for saying sent in sentence today which has been said so um and and I should say Brett is also leading a you know the projects on on that side which we didn't get chance to talk about about the about the nomenclature and all that that happens with the semantic of sentence and Consciousness and agency and all that um so um and and and and and and the whole ethic side of thing so it's they're big big uh opportunities and courses open uh open challenges and and I think you yeah like actually I do you make a good point I should use this opportunity if I may Daniel at least for two flag yeah we are we are engaging in her paper um so if you are interested in the words that we use whether you think that we use them correctly or incorrectly uh we would invite anyone you know we've made these offers before but it's a great place to say it again reach out um we are setting up and we will have a formal invitation uh somewhat soon I can't exactly say when because there's so much going on at the moment but fairly soon we'll have an invitation come join us let's work together and just like figure out what language we want to use as a community because uh then we can move towards actually building building some stuff instead of just being caught up with like what does this word mean and that's going to be necessary so yeah please reach out to us yeah I'll just note one project at The Institute which is the active inference ontology where we've been developing definitions and also translations across languages of core and supplemental terms and we use that to annotate papers we make natural language descriptions of mathematical formalizations and we we totally agree that ontology-based systems engineering for active inference cyber physical systems is how we're going to treat this with the respect and also bring in the discussions around ethics philosophy and so on so this isn't a Linguistics debate that excludes other fields this is like laying down the dance floor so that we can actually have a discussion that's inclusive and meaningful going forward which is what is so important yeah that's awesome I gotta look this up I didn't know that you guys had one of those I'm going to have a look up later I'll uh I'll send it to you in an

email well thank you so much what an awesome line of research and a December surprise or treat for us so super exciting and please always feel welcome to just reach out and join for any other awesome thank you so much Daniel for having us and everyone watching all right bye thank you